

# Exam AI-901: Microsoft Azure AI Fundamentals

Microsoft Press Video Course | 16 lessons | 8 hours runtime | 30 minutes per lesson

## Lesson 1: Identify AI Workloads and Common Scenarios (30 min)

Maps to exam objectives: 1.3.1

- Identify scenarios for common AI workloads, including generative and agentic AI, text analysis, speech, computer vision, and information extraction
- Describe how Microsoft Foundry supports AI workloads in Azure and where each workload type fits in modern cloud architectures
- Differentiate between classic machine learning, generative AI, and agentic AI and map real-world business scenarios to the correct approach

## Lesson 2: How Generative AI Models Work and How to Choose Them (30 min)

Maps to exam objectives: 1.2.1 - 1.2.3

- Describe how generative AI models work
- Identify an appropriate AI model, based on capabilities
- Identify appropriate model deployment options and configuration parameters

## Lesson 3: Responsible AI: Fairness, Reliability and Safety, Privacy and Security (30 min)

Maps to exam objectives: 1.1.1 - 1.1.3

- Describe considerations for fairness in an AI solution
- Describe considerations for reliability and safety in an AI solution
- Describe considerations for privacy and security in an AI solution

## Lesson 4: Responsible AI: Inclusiveness, Transparency, Accountability (30 min)

Maps to exam objectives: 1.1.4 - 1.1.6

- Describe considerations for inclusiveness in an AI solution
- Describe considerations for transparency in an AI solution
- Describe considerations for accountability in an AI solution

## Lesson 5: Text Analysis and Speech Concepts (30 min)

Maps to exam objectives: 1.3.2, 1.3.3

- Describe common text analysis techniques, including keyword extraction, entity detection, sentiment analysis, and summarization
- Identify features and capabilities of speech recognition and speech synthesis
- Describe how natural language processing powers both classic text analysis and modern generative language models

## Lesson 6: Computer Vision and Image-Generation Concepts (30 min)

Maps to exam objectives: 1.3.4

- Identify features and capabilities of computer vision and image-generation models
- Describe core computer vision tasks, including image classification, object detection, OCR, and facial analysis
- Describe how multimodal and image-generation models interpret visual input and produce visual outputs from natural-language prompts

## Lesson 7: Information Extraction Concepts (30 min)

Maps to exam objectives: 1.3.5

- Identify techniques to extract information from text, images, audio, and videos
- Describe how multimodal models and Azure Content Understanding convert unstructured content into structured, grounded representations
- Compare rule-based, machine learning, and multimodal approaches to information extraction and when each is appropriate

## Lesson 8: Tour Microsoft Foundry and Deploy Your First Model (30 min)

Maps to exam objectives: 2.1.2

- Deploy a model and interact with it in the Foundry portal
- Navigate the Microsoft Foundry portal and identify its primary workspaces, projects, model catalog, and playground surfaces
- Select a model from the Foundry model catalog based on task requirements, context length, and cost footprint

## Lesson 9: Craft Effective System and User Prompts (30 min)

Maps to exam objectives: 2.1.1

- Create effective system and user prompts for generative AI models
- Distinguish between system messages and user messages and describe how each shapes model behavior and response quality
- Apply prompt engineering techniques including role instructions, few-shot examples, and structured output formatting

## Lesson 10: Build a Lightweight Chat Client Application with the Foundry SDK (30 min)

Maps to exam objectives: 2.1.3

- Create a lightweight chat client application by using the Foundry SDK
- Authenticate a Foundry SDK client application by using Microsoft Entra identity and keyless credentials
- Send chat completions to a deployed Foundry model from a Python client application and handle the response

## Lesson 11: Create and Test a Single-Agent Solution in the Foundry Portal (30 min)

Maps to exam objectives: 2.1.4

- Create and test a single-agent solution in the Foundry portal
- Describe the components of a Microsoft Foundry agent, including role, instructions, tools, and knowledge sources
- Test an agent in the Foundry portal playground and review the conversation trace to troubleshoot behavior

## Lesson 12: Build a Lightweight Client Application for an Agent (30 min)

Maps to exam objectives: 2.1.5

- Create a lightweight client application for an agent
- Connect a client application to a Foundry agent endpoint by using the Foundry SDK
- Send user messages to an agent and handle multi-turn conversation state from the client application

## Lesson 13: Build a Text Analysis Application with Foundry (30 min)

Maps to exam objectives: 2.2.1

- Build a lightweight application that includes text analysis
- Call Azure AI Language capabilities from a client application by using the Foundry SDK
- Extract entities, key phrases, sentiment, and summaries from unstructured text and display the results in a client UI

## Lesson 14: Build a Speech-Enabled Application with Azure Speech in Foundry Tools (30 min)

Maps to exam objectives: 2.2.2, 2.2.3

- Respond to spoken prompts by using a deployed multimodal model
- Build a lightweight application by using Azure Speech in Foundry Tools
- Configure speech-to-text and text-to-speech in a client application and wire it to a Foundry model for conversational interaction

## Lesson 15: Build a Computer Vision and Image-Generation Application (30 min)

Maps to exam objectives: 2.3.1 - 2.3.3

- Interpret visual input in prompts by using a deployed multimodal model
- Create new visual outputs by using generative models
- Build a lightweight application that includes vision capabilities

## Lesson 16: Extract Information from Documents, Images, Audio, and Video with Content Understanding (30 min)

Maps to exam objectives: 2.4.1 - 2.4.4

- Extract information from documents and forms by using Azure Content Understanding in Foundry Tools
- Extract information from images by using Content Understanding
- Extract information from audio and video by using Content Understanding
- Build a lightweight application with information extraction capabilities by using Content Understanding